General Mathematics — Exercise Sheet 26

Continuous integrands and the fundamental theorem

Variant: Statements

Exercise 1 (Two moving endpoints). Let I and J be nonempty open real intervals. Let $f : J \to \mathbb{R}$ be continuous, and let $\alpha, \beta : I \to J$ be differentiable. Define

$$H(x) = \int_{\alpha(x)}^{\beta(x)} f(t)\,\mathrm{d}t \qquad (x \in I).$$

(a) Use Lecture 26 and the chain rule from Lecture 8 to prove $H'(x) = f(\beta(x))\beta'(x) - f(\alpha(x))\alpha'(x)$.

(b) Apply the formula to $I = J = \mathbb{R}$, $f(t) = 1/(1+t^4)$, $\alpha(x) = x$ and $\beta(x) = x^2$. Compute $H'(x)$ and $H'(0)$.

The integral uses the reversal convention when $\beta(x) < \alpha(x)$.

Exercise 2 (Even and odd integrands). Fix $r > 0$ and let $f : [-r, r] \to \mathbb{R}$ be continuous. The function is *even* if $f(-x) = f(x)$ for every $x \in [-r, r]$. It is *odd* if $f(-x) = -f(x)$ for every such x.

(a) Prove that the integral over $[-r, r]$ equals twice the integral over $[0, r]$ when f is even, and equals zero when f is odd. Use Lecture 26's definite substitution and Lecture 25's linearity/additivity.

(b) Evaluate $\int_{-1}^{1} (x^7 + 4x^4 - 2x^2 + 3)\,\mathrm{d}x$ using these rules.

Exercise 3 (Why the mean-value conclusion needs continuity). Construct a bounded nondecreasing function $h : [0, 2] \to \mathbb{R}$ whose integral average

$$\frac{1}{2}\int_0^2 h(t)\,\mathrm{d}t$$

is not equal to $h(c)$ for any $c \in [0, 2]$. Prove integrability and compute that average. You may use Sheet 25's proved fact that changing finitely many values of an integrable function changes neither integrability nor its integral.

Exercise 4 (Parts with an accumulation integral). Let $f(t) = 1/(1+t^4)$ on $\mathbb{R}$. Define its accumulation integral by

$$A(x) = \int_0^x f(t)\,\mathrm{d}t \qquad (x \in \mathbb{R}).$$

Using Lecture 26, prove

$$\int_0^1 A(x)\,\mathrm{d}x = \int_0^1 \frac{1-x}{1+x^4}\,\mathrm{d}x.$$

Deduce the bounds $1/4 \le \int_0^1 A(x)\,\mathrm{d}x \le 1/2$. The task does not require a closed formula for A.